

Dog Shelter Management System Software Design Document

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Revision History

Revision #	Change Description	Revision Date	Revised By	Approved By
1.0	Initial system design including overall architecture and core feature outline (Dog Intake, Medical Records, Adoption, Status Tracking)	05/06/2026		

Revision #	Change Description	Revision Date	Revised By	Approved By
1.1	Added design details for Dog Intake System and Dog Status Tracking, including data flow and component interactions	05/07/2026		
1.2	Added design details for Medical Records System and Adoption Management System, including expanded architecture and workflows	05/08/2026		
1.3	Finalized system design with refinements, improved consistency, and inclusion of future enhancement	05/09/2026		

1 Introduction

1.1 Purpose

This Software Design Document (SDD) describes the architecture, design decisions, and system components for the Dog Shelter Management System. It is intended to guide the development team throughout all project phases and provide a technical reference for stakeholders.

1.2 Intended Audience

This document is intended for:

- Software Developers and Project Managers
- Shelter Administrators and Staff
- Users (Potential Adopters) and Testers

1.3 System Overview

The Dog Shelter Management System is a web-based application designed to help animal shelters streamline and manage their day-to-day operations in an organized and efficient manner. The system is built around five core features that together cover the full lifecycle of a dog's stay at the shelter:

1. **Dog Intake System** - Add new dogs and store basic information (name, breed, age, arrival date)
2. **Medical Records System** – Track vaccinations, store medical history, and update health status
3. **Adoption Management System** – Allow users to submit adoption applications and staff to review, approve, or reject them
4. **Dog Status Tracking** – Manage and display availability statuses: Available, Pending, or Adopted
5. **Admin Dashboard** – Provide staff with a centralized interface to manage dogs, medical records, and applications

1.4 Scope

This document covers the initial design and setup of the system as defined in the Start Objective. Subsequent versions of this document will reflect updates made at each project checkpoint.

2 System Architecture

2.1 High-Level Architecture

The Dog Shelter Management System follows a standard three-tier architecture:

- **Frontend (Presentation Layer):** A web-based user interface accessible by both general users and shelter staff.
- **Backend (Application Layer):** A RESTful API server that handles business logic, data processing, and communication between layers.
- **Database (Data Layer):** A relational database that persists all dog, medical, adoption, and user data.

2.2 Workflow

At a high level, users interact with the frontend to perform actions such as viewing available dogs or submitting adoption applications. The frontend communicates with the backend API, which processes requests and reads from or writes to the database. Staff and administrators access additional management views through the Admin Dashboard feature. When an adoption application is submitted, the backend evaluates the request and updates the dog's status accordingly (Pending on approval, Available on rejection).

2.3 Data Flow

The diagram below illustrates how data travels through the system when a user or staff member performs an action. Using dog intake as an example, the flow proceeds from form submission through the backend API for validation, into the database for storage, and finally back to the UI as a refreshed dog listing.



Figure 1: Data flow diagram – dog intake example (form → backend → database → UI)

2.4 Component Interaction

The diagram below shows how the three system layers communicate with each other. The frontend sends HTTP requests to the backend API and receives JSON responses. The backend API issues SQL queries to the database and processes the returned result sets.

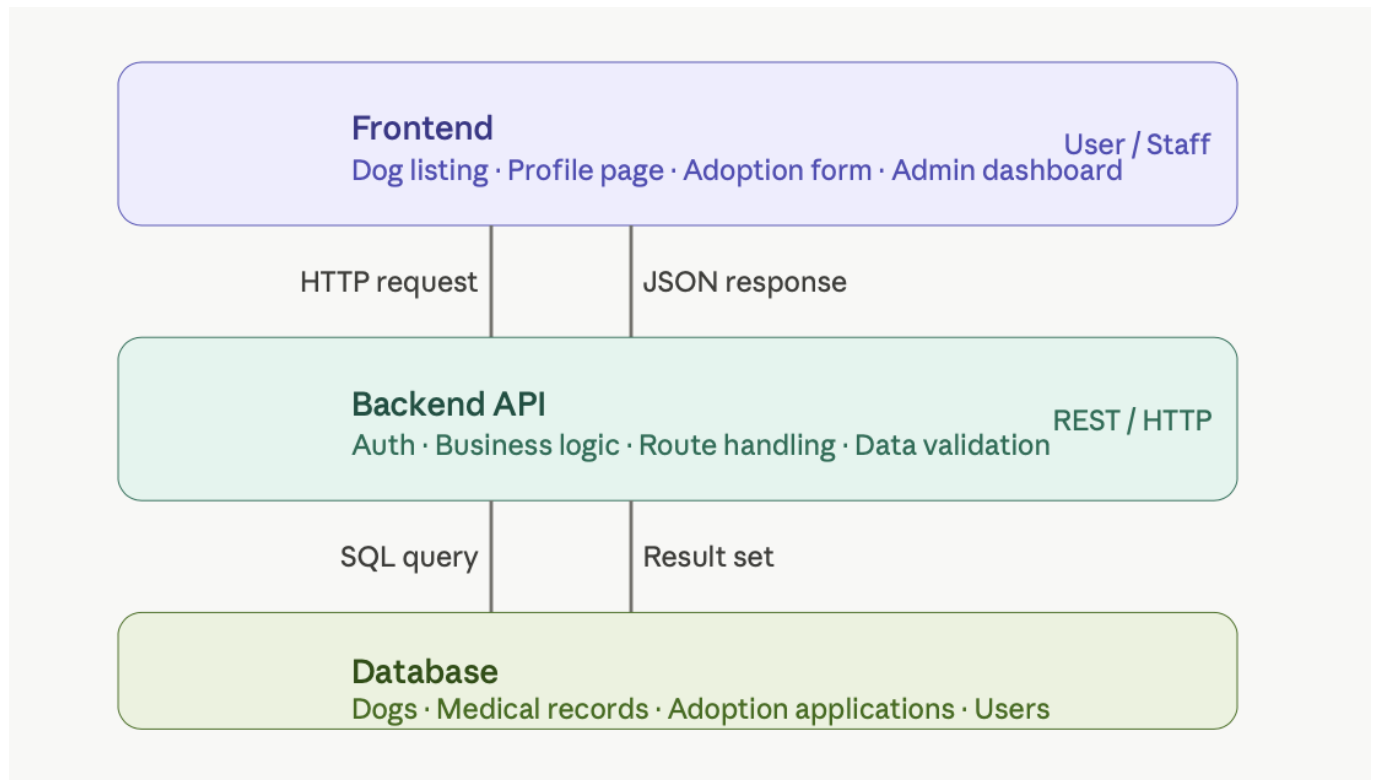


Figure 2: Component interaction diagram – frontend ↔ backend ↔ database

2.5 Technology Stack

Layer	Technology
Frontend	docker-compose.yml
Backend	Node.js
Database	MySQL
Containerization	Docker
Version Control	GitHub

3 User Interface

3.1 Overview

The user interface will consist of two primary user roles. General users (potential adopters) can browse available dogs, view individual dog profiles, and submit adoption applications. Staff and administrators have access to all general user functionality, plus additional management tools including the dog intake form, medical records editor, adoption application reviewer, and the Admin Dashboard.

3.2 Key Pages

- **Home / Dog Listing Page:** Displays all dogs with their current status (Available, Pending, Adopted) with filtering and search options.
- **Dog Profile Page:** Shows individual dog details including basic info and health status.
- **Adoption Application Page:** Form for users to submit an adoption application for a specific dog.
- **Admin Dashboard:** Staff-only view for managing all system data including dogs, records, and applications.
- **Dog Intake Form:** Staff-only form to register a new dog into the system.
- **Medical Records Page:** Staff-only interface to view and update a dog's medical history and vaccination records.

Note: Detailed UI mockups and database schema will be added in subsequent versions.

3.3 Database Explanation

The system uses a relational database to store and manage all persistent data. Each core feature of the application maps directly to one or more database tables:

- **Dogs table** – Stores each dog's basic information including name, breed, age, arrival date, and current status (Available, Pending, or Adopted).
- **Medical Records table** – Stores vaccination history, health status updates, and medical notes linked to a specific dog by ID.
- **Adoption Applications table** – Stores all submitted adoption applications, including the applicant's information, the dog requested, and the application status (Pending, Approved, or Rejected).
- **Users table** – Stores user accounts and roles, distinguishing between general users (adopters) and staff/administrators.

When a user or staff member performs an action on the UI – such as adding a dog, updating medical records, or submitting an adoption application – the backend API translates that action into a database operation (INSERT, UPDATE, or SELECT) and returns the result to the UI as a JSON response.

4 Glossary

Term / Acronym	Definition
SDD	Software Design Document
SRS	Software Requirements Specification
API	Application Programming Interface
REST	Representational State Transfer
UI	User Interface
UX	User Experience
Admin	Administrator; a staff member with elevated system permissions
Intake	The process of registering a new dog into the shelter system
Status	A dog's current adoption state: Available, Pending, or Adopted
CRUD	Create, Read, Update, Delete – basic data operations
HTTP	Hypertext Transfer Protocol – used for frontend-backend communication
SQL	Structured Query Language – used for database operations

5 References

- CS3338 Final Project Assignment Document – Professor's Guidelines: <https://calstatela.instructure.com/courses/122817/assignments/2390947>
- Ascent Senior Design Projects: <https://ascent.cysun.org/project/project>
- Node.js Documentation: <https://nodejs.org/en/docs>
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